UNIVERSITY OF ESSEX

Postgraduate Examinations 2017

MACHINE LEARNING AND DATA MINING

Time allowed: **TWO** hours

Candidates are permitted to bring into the examination room:
Calculator – Casio FX-83GT PLUS or Casio FX-85GT PLUS only

The paper consists of **THREE** questions.

Candidates must answer **ALL** questions.

The questions are **NOT** of equal weight.

Questions 1 and 2 are worth 35% of the marks for the paper each.
Question 3 is worth 30% of the marks for the paper.

The percentages shown in brackets provide an indication of the proportion of the total marks for the **PAPER** that will be allocated for that part of the question.

Please do not leave your seat unless you are given permission by an invigilator.
Do not communicate in any way with any other candidate in the examination room.
Do not open the question paper until told to do so.
All answers must be written in the answer book(s) provided.
All rough work must be written in the answer book(s) provided. A line should be drawn through any rough work to indicate to the examiner that it is not part of the work to be marked.
At the end of the examination, remain seated until your answer book(s) have been collected and you have been told you may leave.

Question 1

(a) Support vector classifiers can be thought of as instance-based learners. Explain why, indicate what similarity measures they use, explain what instances they retain and how these contribute to the class prediction. [8%]

(b) In the context of Machine Learning, describe the concept of "linear separability of a set of examples". Draw a sketch of a hypothetical training set in the following three situations: linearly separable case; non-linearly separable case because of noise in the data; intrinsically non-linearly separable case. [5%]

(c) Figure 1.1 (below) shows a set of training examples and the decision function of a non-linear support vector classifier that was trained on these data. Identify which of the training points are support vectors and why. [5%]

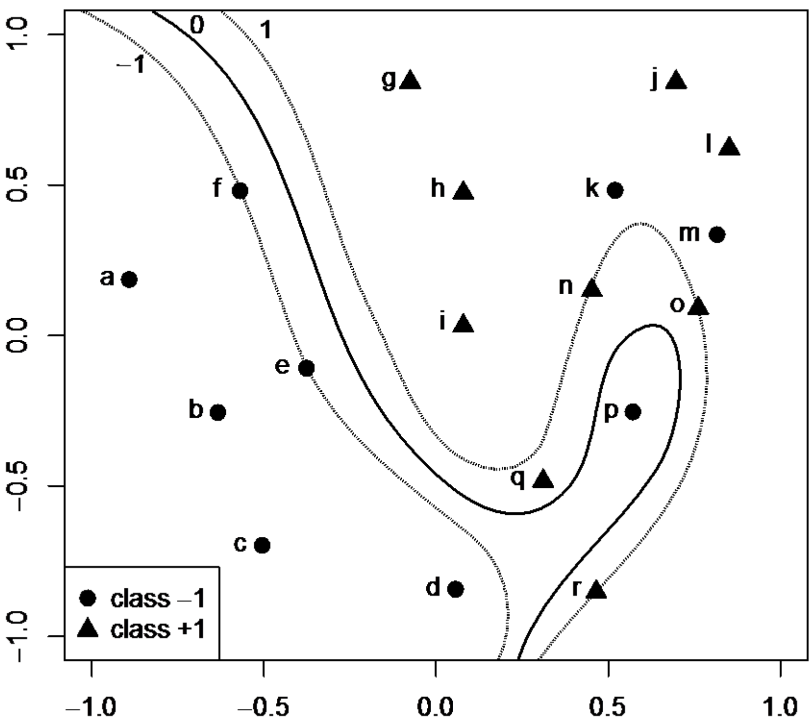

Figure 1.1

Question 1 continues…

<u>Question 1</u> (continued)

(d) A support vector classifier with a radial basis function kernel ($\gamma = 0.4$) was trained on some data. Out of all training instances, the algorithm selected $S = \{2,5,8\}$ as support vectors: [9%]

$x^{(2)} = [0.2,0.8,0.1]$ with label $r^{(2)} = +1$ and optimal weight $\alpha_2 = 0.25$,

$x^{(5)} = [-0.5,0.2,0.7]$ with label $r^{(5)} = -1$ and optimal weight $\alpha_5 = 1.0$,

$x^{(8)} = [0.5,0.8,0.1]$ with label $r^{(8)} = +1$ and optimal weight $\alpha_8 = 0.5$.

The optimal bias of the classifier is $b = 0.3$. Compute the decision function $h(x)$ and the class of the new example $x = [-0.5,0.8,0.1]$.

You may find it helpful to remember that:

$K_{RBF}(x,z) = \exp(-\gamma \|x-z\|^2)$,

$h(x) = [\Sigma_{s \in S} \alpha_s r^{(s)} K(x^{(s)},x)] - b$.

(e) Draw a sketch of the *0-1 loss* and of the *hinge loss* functions. Briefly explain three disadvantages of the *0-1 loss* compared to the *hinge loss* when building linear classifiers. [8%]

Question 2

A movie streaming website would like to offer discounted prices to customers, based on their personal preferences. They extract the titles watched by five customers and would like to carry on some preliminary analysis:

Viewer 1:	Aliens, Inception, Avatar
Viewer 2:	Aliens, Titanic, Scarface, Goodfellas, Avatar
Viewer 3:	Aliens, Inception, Goodfellas, Avatar
Viewer 4:	Aliens, Goodfellas, Scarface
Viewer 5:	Aliens, Inception, Avatar

(a) Association rules are used to discover regularities in transaction data. By applying the Apriori Algorithm, find all the frequent itemsets for this data. Use a minimum support threshold of 60% (i.e. a minimum support count of 3). [8%]

(b) The website wishes to increase the number of views of "Inception". Using your answers to part (a) of this question, derive all the association rules that predict a sale for "Inception", state their confidence, and calculate their lift. [10%]

(c) The website would also like to categorise the movies to make the website easier to navigate. For the purpose of this exercise, the similarity of two movies is defined as the number of viewers who have watched both movies. The similarity of two groups of movies is defined as the similarity of the *least* similar pair of movies where each member of the pair is drawn from a different group. Construct a table showing the similarities between each pair of movies among the six listed above. Hence draw the dendrogram that would be produced using an agglomerative hierarchical clustering technique. [14%]

(d) Suppose that after completing the dendrogram you are provided with an additional transaction; could a revised dendrogram be constructed by modifying the one that has already been constructed? [3%]

<u>Question 3</u>

(a) Given a particular data set, overfitting is less likely to be a problem using naïve Bayes learning than with decision tree induction. Discuss briefly why this is. [6%]

(b) Suppose you invent a new classifier made of a shallow decision tree with a naïve Bayes classifier at each leaf. To make a prediction based on m features your classifier traverses the tree depending on the value of k features (where $k < m$ is the depth of the tree, assumed fixed) and then applies the corresponding naïve Bayes classifier, which uses the $m-k$ features not already employed on the specific path from the root to that leaf. Design a simple learning algorithm for training this type of classifier. You are encouraged to use a practical example to explain each step of the algorithm. [14%]

(c) Using your answer to part (a), discuss why the algorithm designed in part (b) may work better than both a standard decision tree and a standard naïve Bayes on the same data. In particular, discuss some characteristics of a dataset for which this algorithm may be particularly suited. [10%]

END OF PAPER CE802-7-AU